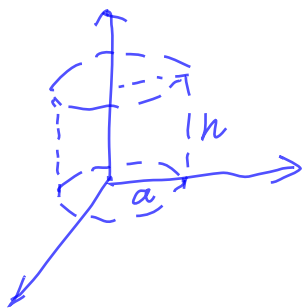


## Quiz 11

**Instructions:** There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

**Problem 1.** (8 points) Find the scalar surface integral  $\iint_{\mathbf{S}} (x^2 + y^2) dS$ , where  $\mathbf{S}$  is the lateral surface of the cylinder with radius  $a$  and height  $h$  whose axis is the  $z$ -axis.



$$\mathbf{r} = \Phi(s, t) = (a \cos s, a \sin s, t) \\ 0 \leq s \leq 2\pi, \quad 0 \leq t \leq h.$$

$$\mathbf{T}_s = \frac{\partial \Phi}{\partial s} = (-a \sin s, a \cos s, 0)$$

$$\mathbf{T}_t = \frac{\partial \Phi}{\partial t} = (0, 0, 1)$$

$$\mathbf{N}(s, t) = \mathbf{T}_s \times \mathbf{T}_t = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -a \sin s & a \cos s & 0 \\ 0 & 0 & 1 \end{vmatrix} \\ = (a \cos s) \vec{i} + (a \sin s) \vec{j}$$

$$\Rightarrow \|\mathbf{N}(s, t)\| = \sqrt{(a \cos s)^2 + (a \sin s)^2} = a.$$

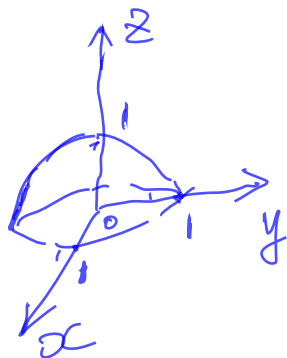
$$\iint_{\mathbf{S}} (x^2 + y^2) dS = \iint_{\Phi} (x^2 + y^2) dS \\ = \int_0^h \int_0^{2\pi} (x^2 + y^2) \|\mathbf{N}(s, t)\| ds dt \\ = \int_0^h \int_0^{2\pi} a^3 ds dt \\ = 2\pi a^3 h$$

**Problem 2.** (12 points) Let  $\mathbf{S}$  be the portion of the surface  $z = 1 - x^2 - y^2$  that lies above the  $xy$ -plane and suppose  $\mathbf{S}$  is oriented up (i.e., the standard normal vector  $\mathbf{N}$  faces up).

Find the flux of the vector field  $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$  across  $\mathbf{S}$ , i.e.,  $\iint_{\mathbf{S}} \mathbf{F} \cdot d\mathbf{S}$ .

$$\mathbf{S}: \Phi(x, y) = (x, y, 1 - x^2 - y^2), \quad (x, y) \in D$$

$$\text{where } D = \{(x, y) \mid x^2 + y^2 \leq 1\}$$



$$\begin{aligned} \Rightarrow \mathbf{N}(x, y) &= \mathbf{T}_x \times \mathbf{T}_y \\ &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & -2x \\ 0 & 1 & -2y \end{vmatrix} = (2x, 2y, 1) \end{aligned}$$

$\downarrow$   
 consistent with the  
 orientation of  $\mathbf{S}$

Thus

$$\begin{aligned} \iint_{\mathbf{S}} \mathbf{F} \cdot d\mathbf{\vec{S}} &= \iint_{\Phi} \mathbf{F} \cdot d\mathbf{\vec{S}} = \iint_D \mathbf{F}(\Phi(x, y)) \cdot \mathbf{N}(x, y) dx dy \\ &= \iint_D (x, y, z) \cdot (2x, 2y, 1) dx dy \\ &= \iint_D (x, y, 1 - x^2 - y^2) \cdot (2x, 2y, 1) dx dy \\ &= \iint_D (x^2 + y^2 + 1) dx dy \quad \leftarrow \begin{array}{l} x = r \cos \theta \\ y = r \sin \theta \\ 0 \leq r \leq 1 \\ 0 \leq \theta \leq 2\pi \end{array} \\ &= \int_0^1 \int_0^{2\pi} (r^2 + 1) r d\theta dr \\ &= \int_0^1 2\pi(r^3 + r) dr \\ &= 2\pi \left[ \frac{r^4}{4} + \frac{r^2}{2} \right]_{r=0}^{r=1} = \frac{3\pi}{2} \end{aligned}$$